

EC2 · EBS · ENI

What one server is made of, and how it connects



What is one server

Rent via EC2, clone from an AMI.

What stays, what disappears

Stop/terminate, EBS, snapshots.

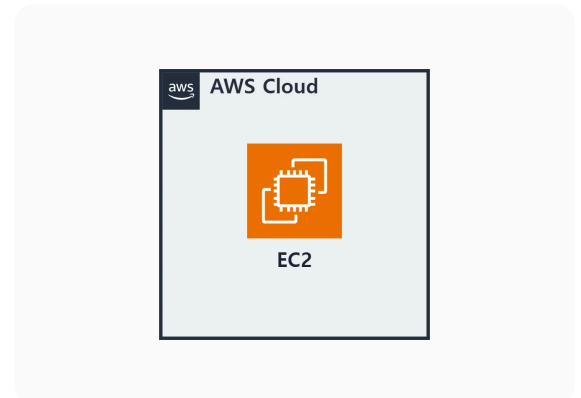
How it connects

One ENI holds IPs and SGs.

01 EC2 — AWS servers for rent

Elastic Compute Cloud

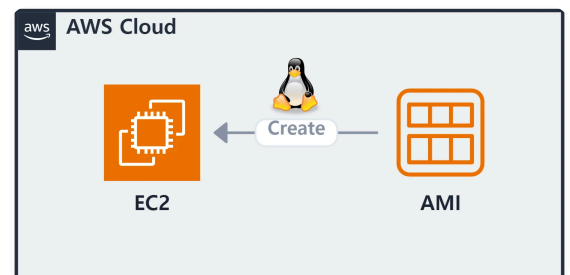
- Renting a server from AWS
- A **virtual server** you start when needed and stop when not.
- Easy to scale up and down, hence **Elastic**. Two C's make **EC2**



02 AMI — a server template

Amazon Machine Image

- The **template** that sets the OS and settings to launch with.
- A configured server can be baked back into an AMI.
- Then launch as many identical servers as you like.



Seven choices at launch

In the console wizard's order

OS	Linux · Windows · macOS — set by the AMI	CPU	How many vCPUs — set by the type's family and size
Memory	RAM. Same size, different amount per family	Storage	Network-attached EBS · EFS, or instance store on the physical host
Network	NIC bandwidth, and whether to assign a public IP	Firewall	Security group — who may connect, on which ports
First-boot script	User data — a script that runs once, at first boot		

03 Reading type names

One short name is a **spec sheet**

t3.micro

t Family — T-M general, C compute, R memory

3 Generation — higher is newer, usually cheaper and faster

micro Size — each step doubles memory

Family first, then generation and size.

Instance families

Exam point

Family	Use	When to pick	Example
T	General · burstable	Mostly idle web / dev servers	t3.micro
M	General purpose	Balanced CPU and memory; the default	m7g.large
C	Compute optimized	Compute-heavy batch · game servers	c7i.xlarge
R / X	Memory optimized	Caches · in-memory DBs	r7g.large
I / D	Storage optimized	High IOPS · large local disks	i4i.large
G / P	Accelerated computing	GPU training · inference · rendering	g5.xlarge

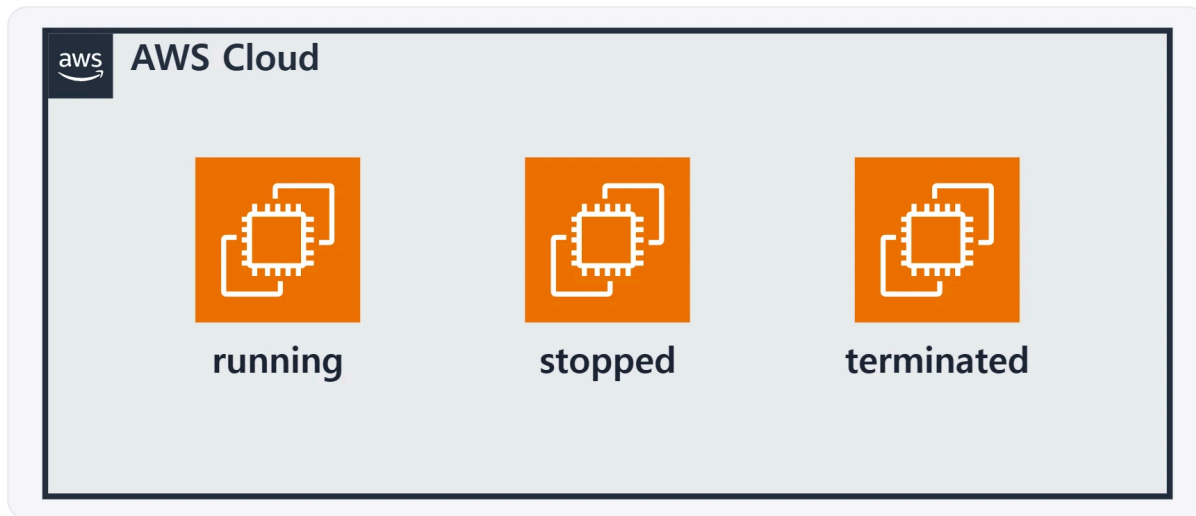
Instance sizes

nano → micro → small	Smallest three: dev, test
xlarge → 2xlarge → 4xlarge...	2× memory, and more vCPUs
.metal — bare metal	Whole physical server, no hypervisor. (For licenses tied to hardware)

medium → large	Where production starts
a · g · i · n suffixes	AMD · Graviton · Intel · enhanced networking

04 Three states — where billing splits

You pay for the instance only while it runs.



running

On. Being billed.

stopped

Off. Instance charges stop.

terminated

Deleted for good. No undo.

State transitions

Exam point

pending

starting up



running

billed



stopping

shutting down



stopped

no instance fee



terminated

no way back

A reboot doesn't change state. It's a restart, so the public IP and instance store stay. Hibernate saves memory to the root volume, stops, then resumes where it left off.

Placement groups — how instances sit physically

Cluster	Packed into one rack. Lowest latency; if the rack fails, they all fail
Spread	Scattered across distinct hardware. Limited instances per AZ
Partition	Split by groups of racks. For distributed stores like HDFS-Cassandra

 Pricing models

On-Demand

Pay per second for what you use.
The default, and the priciest.

Savings Plans

Commit to a fixed \$/h for 1–3 years, get a discount.

Reserved

Reserve a specific type for 1–3 years. You commit to instances, not dollars.

Spot

Spare capacity, up to 90% off.
Reclaimed after a 2-minute notice.

Dedicated Host

A whole physical server. For licenses tied to cores.

Capacity Reservation

Holds capacity in advance. It secures a spot; it saves nothing.

 Which one to choose

Servers running 24/7 **Savings Plans** — commit to an hourly spend for a discount

Interruptible jobs **Spot** — up to 90% off, reclaimed after a 2-minute notice

Short trial runs **On-Demand** — no commitment. Priciest, and most flexible

Per-core licenses **Dedicated Host** — rent the whole physical server

 What you pay for

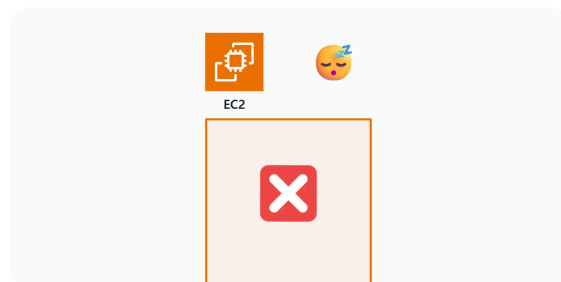
Stopping halts only the first row

—	When charged	To stop it
Instance hours	Only while running	Stop the instance
EBS volume	As long as it exists (GB-month)	Delete the volume. Stopping won't do it
Snapshot	As long as it's kept (GB-month)	Delete it
Public IPv4	From allocation on (hourly)	Release unused EIPs. Billed even while stopped
Data transfer	Outbound (per GB)	Inbound is free; private-IP traffic within one AZ is free

05 Instance Store

What's lost when you stop

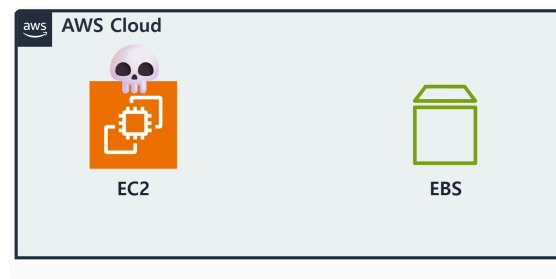
- Data on an EC2 **instance store** is lost when you stop.
- A temporary disk on the physical host, so it's fast
- It survives a **reboot**, but not a stop or terminate.



06 EBS — over the network

Elastic Block Store

- It outlives the instance.
- Detach it and **attach it to another instance**.
- Own lifetime, own bill (billed even with the instance off)



👤 Stop (stopped) vs terminate (terminated)

[Exam point](#)

—	Stop	Terminate
Instance charge	Stops	Stops
EBS volume	Still billed	Root volume deleted
Data on it	Kept	Lost
Public IP	Changes (EIP excepted)	Released
Undo	Start it again	Not possible

🗄 Three disks, easily confused

EBS

A network-attached disk. Survives instance deletion; detach it and attach it to another instance.

Instance store

A temporary disk on the physical host. Fast, but lost on stop or terminate. Survives a reboot.

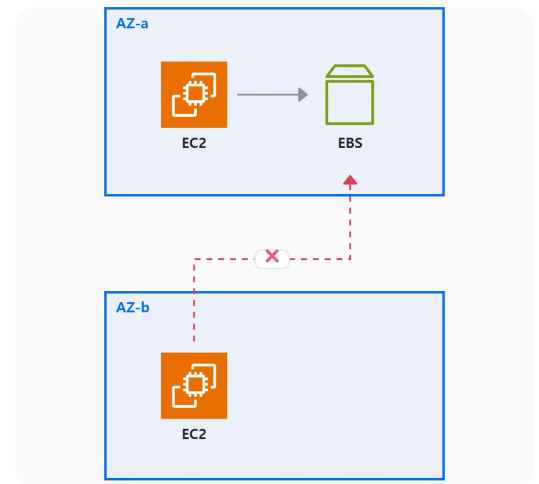
Snapshot

Incremental EBS backup in S3. The start for moving AZ/Region or baking an AMI.

07 A volume lives in one AZ

It **can't attach** to an instance in another AZ.

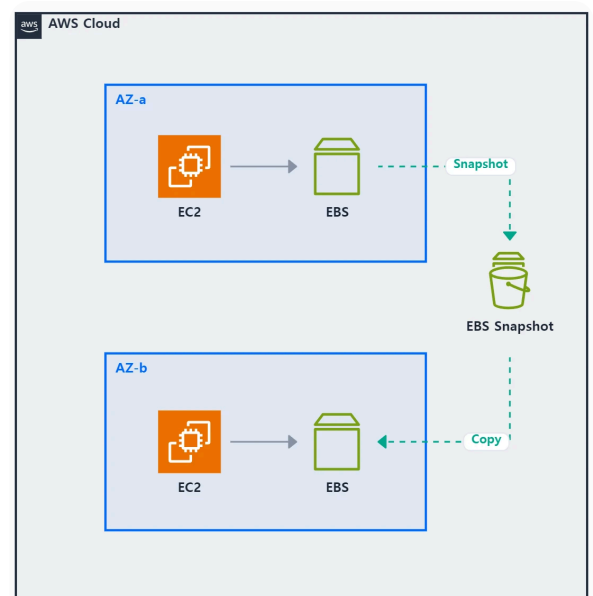
- Volume and instance must be in the same AZ.
- So changing AZ is a **copy**, not a move.
- For Region-wide storage, use another service such as EFS.



08 Snapshots — how to cross AZs and Regions

Take a snapshot and **create a new volume in another AZ or Region**.

- Stored in S3 **incrementally**. Only the first one is full
- A volume made from a snapshot can be larger than the original.
- Baking an AMI starts from this snapshot, too.



What snapshots do

Exam point

Incremental backup

Stored in S3; only the first one is full.

Moving AZ · Region

Copy it, then create the volume in the other AZ or Region. (The only way a volume crosses AZs)

Where AMIs start

Baking an instance into an AMI comes down to snapshotting its root volume.

Automation and retention

DLM takes and deletes them on a schedule. Old ones move to the archive tier.



Encryption — decide at creation

With a KMS key

Turn it on when you create the volume. Data, snapshots and volumes made from them are all encrypted.

Can't enable it later

An existing volume must go **snapshot → encrypted copy → new volume**.

No performance cost

No noticeable penalty, so better to leave it on by default.

Volume rules — where they attach, when they go

Exam point

AZ A volume belongs to **one AZ**. It attaches only to instances there

Lifetime Independent of the instance — detach it and attach it elsewhere

While stopped Volumes remain and are **still billed**

Attachment One volume, one instance (except io1·io2 **Multi-Attach**)

Root volume Delete on termination is **on by default**. Off for data volumes

EBS volume types — which to pick

Root volumes: SSD only

Type	What	When to pick
gp3	General Purpose SSD (default)	Baseline 3,000 IOPS · 125 MB/s. Size and speed set separately
gp2	General Purpose SSD (older)	3 IOPS per GiB — size sets performance
io2	Provisioned IOPS SSD	You buy the IOPS. Latency-sensitive DBs
st1	Throughput Optimized HDD	Large sequential files. Logs · big data
sc1	Cold HDD	Cheapest. Rarely read data

What you can change later

Only upward

Size — grow only

It grows while in use, but **can't shrink**. To shrink, create a new volume and copy the data over.

Type — changeable

Change it live, e.g. gp2 → gp3. Performance takes a while to settle afterwards.

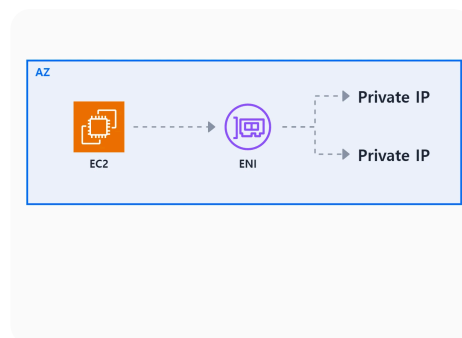
Limits

Volumes 1 GiB – 16 TiB; gp3 up to 16,000 IOPS · 1,000 MB/s. Beyond that, the instance's own EBS bandwidth is the cap.

09 ENI — EC2's virtual NIC

Elastic Network Interface

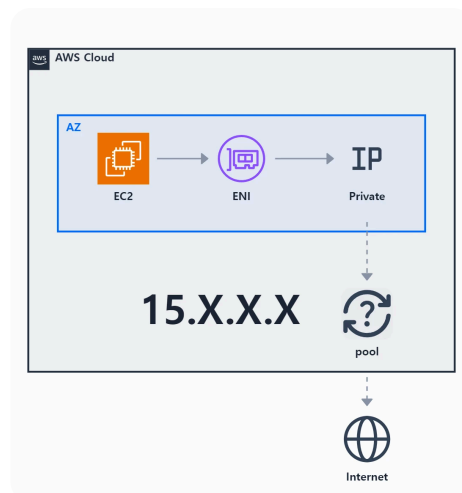
- An ENI also lives in **one Availability Zone** — same rule as volumes.
- Each instance has one primary ENI; its instance type may allow more.
- Detach it and move it to another instance — the addresses go with it.
- IPs, MAC and security groups belong to **this card**, not the instance.



10 Where a public IP attaches

Not to the instance, but **to the ENI's private IP**.

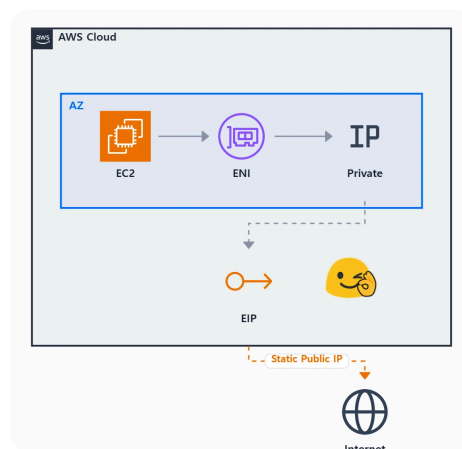
- Assigned **at random** from AWS's pool at launch — you can't choose it.
- The private IP is fixed for the instance's life
- Since 2024-02, public IPv4 is **billed hourly**, always.
- Stop and start the instance, and the public IP **changes**.



11 EIP — a fixed address

Elastic IP

- A **static public IP** allocated to your account and attached to an ENI.
- Billed hourly, attached or not. Release it when unused.
- Belongs to a Region. 5 per account by default
- The other way to pin an address is a domain (Route 53).

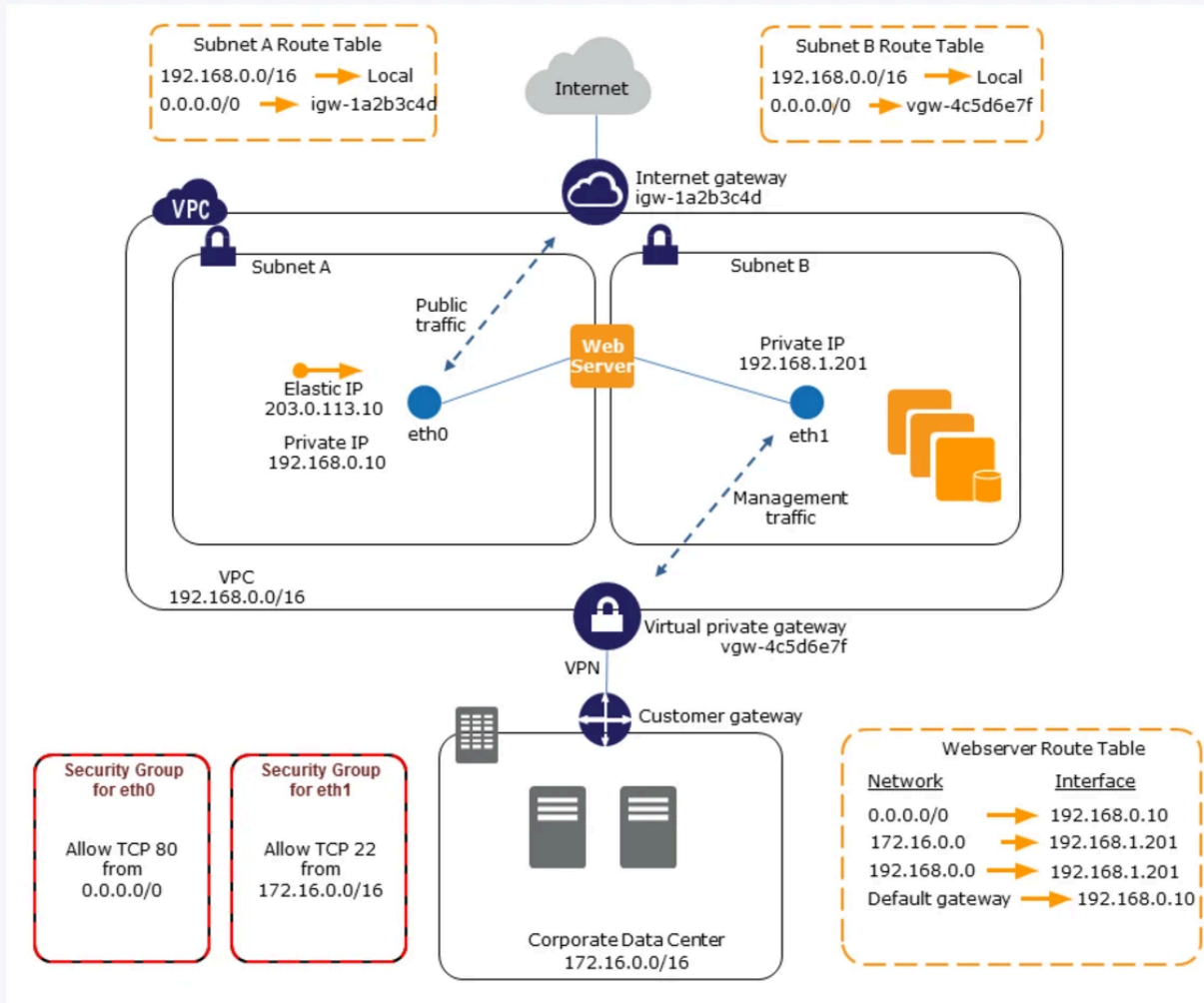


ENI rules

Exam point

AZ	An ENI also belongs to one AZ
Extra ENIs	As many as the instance type allows.
Security group	Stateful allow rules on the ENI , not the instance

Primary ENI	One per instance (eth0). Can't be deleted; goes with the instance
Private IP	1 per ENI by default; more secondary IPs by type
MAC address	Moves with the card (MAC-bound license? Move the ENI)



eth0 gets its private IP from Subnet A, eth1 from Subnet B; **each card has its own security groups**.

[AWS docs: "Multiple network interfaces"](#) ↗

Four things the diagram shows

Exam point

An ENI belongs to a subnet

Its private IP comes from that subnet's range.

Security groups per card

They attach to the ENI, not the instance — in the diagram eth0 opens 80·443, eth1 only 22.

An EIP sits on a private IP

One public IP per private IP on the card. Move the card and the addresses follow.

One instance, two subnets

In one AZ, cards can span subnets: a public one, plus a private one for admin

IP **Three kinds of address**

Exam point

—	How you get it	After stop/start	Cost
Private IP	Automatic, from the subnet	Unchanged	Free
Public IP	Random from the pool at launch	Changes	\$0.005 per hour
Elastic IP	Allocate to the account, then attach	Unchanged	\$0.005 per hour (even unattached)

 Access and permissions — where accidents happen**Key pair**

You get the private key (.pem) **once**, at creation. Lose it and there's no reissue.

Security group

Allow rules only — who may connect, on which ports. Never open 22/3389 to 0.0.0.0/0.

IAM role

Attach it via an instance profile. Never keep access keys on the server.

 Checklist — before you launch an instance

- ☐ Saved the key pair? It can't be downloaded again
- ☐ Checked the root volume's **Delete on termination** setting?
- ☐ Turned on termination protection?
- ☐ Tagged it (Name, Env, Owner)? Untagged, it's lost in the bill
- ☐ Does the security group keep 22/3389 closed to 0.0.0.0/0?
- ☐ Know that User data runs **once, at first boot**?
- ☐ Attached an IAM role? Don't embed access keys
- ☐ Enabled volume encryption? Only possible **at creation**



Scope — what belongs to a Region, what to an AZ

[Exam point](#)

Resource	Scope	Good to know
AMI	Region	Visible only in its own Region — copy it to use it elsewhere
Snapshot	Region	Copyable across Regions — the only way a volume leaves its Region
EBS volume	AZ	Attaches only to instances in the same AZ
Instance	AZ	Lives in the AZ picked at launch — to move, relaunch from an AMI
EIP	Region	Moves across AZs within its Region. 5 per Region per account by default



Stop first? What works live, what needs a stop

Snapshots need no stop

Task	Instance	Good to know
Take a snapshot	Running	Captures only what's on disk — if unsure, pause first
Resize·retype a volume	Running	Then grow the file system in the OS to use it
Attach/detach data volume	Running	Unmount before detaching
Detach the root volume	Stopped	Can't be detached while running
Change instance type	Stopped	Change it while stopped — the public IP changes too
Edit User data	Stopped	Only while stopped. Still runs only at first boot by default
Bake an AMI	Running	Reboots first by default — skip it and consistency is on you



Recovery — which fix for which loss

The size of the loss sets the method

One file	Leave the volume be; make a new volume from a snapshot, attach it, copy the file out
Whole volume	Restore it from a snapshot. Same AZ: directly; another AZ: copy first
Won't boot	Detach the root volume, attach it to another instance as a data disk, fix it
Whole server	If you made an AMI, launch a new one from it
Terminated	The root volume is gone. Without a snapshot or AMI, there's no way back

 Try these!

Practice

1 You stop a dev server every evening and start it each morning, yet the hours it's off don't cost \$0. Which charges continue while it's stopped? Choose all that apply. [2 answers](#)

- ☐ A Instance-hour charges
- ☐ B The attached root EBS volume
- ☐ C Data left on the instance store
- ☐ D The EIP attached to the instance
- ☐ E The instance's data transfer

2 A fleet that scales between 10 and 50 instances with load, across several Availability Zones, must read and write the same 50 GB folder. What do you attach?

- ☐ A Create one volume and attach it to every instance
- ☐ B Give each instance a volume and keep them in sync
- ☐ C Create an EFS file system and mount it everywhere
- ☐ D Keep it on instance store and copy it as needed

3 A database with heavy random reads and writes. Performance must be consistent, and the data must outlive the instance. Which disk?

- ☐ A Instance store — on the physical host, so the fastest
- ☐ B gp2 — a bigger volume brings more IOPS
- ☐ C st1 — Throughput Optimized HDD
- ☐ D io2 — Provisioned IOPS SSD

Answers and explanations

- 1 **B · D** Stopping halts **only the instance-hour charge**. A volume has its own lifetime, so it stays and is billed as long as it exists. Since 2024-02, public IPv4 and EIPs keep costing money even with the instance off.
- 2 **C** **A volume lives in one AZ and attaches to one instance.** io1-io2 Multi-Attach is the exception, but only for up to 16 Nitro instances in the same AZ — not 50 across AZs. Once many machines write the same data at once, you need a **file system**, not a block device; EFS is Regional and many instances mount it together. Instance store is a disk on the physical host and can't be shared at all.
- 3 **D** Two conditions, each ruling out one option. **Must outlive the instance** rules out instance store — fast, but lost on stop or terminate. **Must be consistent** rules out gp2: at 3 IOPS per GiB its performance is tied to size, and small volumes rely on bursts. st1 is for large sequential files, not random IO. io2 keeps delivering the IOPS set at creation, so it is the answer when "consistent performance" is required.

⚠ Common traps

Exam favorite

Thinking a stopped instance costs \$0

Only instance charges stop. Attached EBS and held public IPv4 are still billed.

Picking the size first

Choose the family (the letter) first. At the same size, C and R are different servers.

Thinking User data runs every boot

It runs only once, at the very first boot.

Attaching a volume from another AZ

EBS lives in one AZ. The only way across is to snapshot and copy.

Hitting Terminate, meaning Stop

Even the root volume is gone. Termination protection is the only safeguard.

Trusting the public IP to stay

It changes on stop/start. For a fixed address, use an Elastic IP or a domain.

Thinking volumes can shrink

They only grow. To shrink, create a new volume and copy the data over.

Blocking with a security group

Security groups have no deny rules, only allows — blocking is the network ACL's job.

📖 Terms in one line

AMI A template for stamping out servers. Holds the OS and initial setup.

Instance store A temporary disk on the physical host. Lost on stop or terminate.

Snapshot An incremental backup of EBS in S3.

Termination protection Blocks the terminate API. Doesn't block stop.

Graviton AWS's own ARM CPU. Marked by a g in the type name.

Availability Zone (AZ) An isolated group of data centers in a Region. EBS and ENIs belong to one.

EIP A static public IP allocated to your account. Billed attached or not.

DLM Lifecycle manager that takes and deletes snapshots on a schedule.

Instance type A spec sheet in one name: family, generation, size.

EBS A network-attached disk. Outlives the instance and can be moved.

User data A script run as root, once, at first boot.

Dedicated Host Rent the physical server itself. For per-socket or per-core licenses.

IMDS Internal-only address where an instance looks up its own metadata (169.254.169.254).

ENI The virtual network card EC2 talks through. Carries IPs, MAC and security groups.

Multi-Attach Attaches one io1·io2 volume to several instances in the same AZ.

In one line

One server is EC2 (compute) · EBS (disk) · ENI (network card). All three are bound to one Availability Zone; to reach beyond it, take a snapshot or pin the address.

EC2

AMI

Instance type

States

EBS

Snapshot

ENI

EIP

 Official docs

AWS guides · video · articles

**EC2 User Guide** [↗](#) [p. 1](#)

The source on instances · AMIs · lifecycle
docs.aws.amazon.com/ec2

**Instance lifecycle** [↗](#) [pp. 3 · 5](#)

What stays and what goes in each state
docs.aws.amazon.com/AWSEC2/latest/UserGuide/ec2-instance-lifecycle.html

**AMI** [↗](#) [p. 1](#)

What an image does and doesn't contain
docs.aws.amazon.com/AWSEC2/latest/UserGuide/AMIs.html

**Instance types** [↗](#) [p. 2](#)

How to read family names; tables by generation
aws.amazon.com/ec2/instance-types

**EBS User Guide** [↗](#) [pp. 5 · 7](#)

All that follows from volumes living outside the instance
docs.aws.amazon.com/ebs/latest/userguide/what-is-ebs.html

**EBS volume types** [↗](#) [p. 7](#)

IOPS and throughput of gp3 · io2 · st1
docs.aws.amazon.com/ebs/latest/userguide/ebs-volume-types.html

**EBS snapshots** [↗](#) [pp. 6 · 11](#)

The only way to move a volume out of its AZ
docs.aws.amazon.com/ebs/latest/userguide/ebs-snapshots.html

**ENI** [↗](#) [pp. 8 · 9](#)

Moving a card to another instance
docs.aws.amazon.com/AWSEC2/latest/UserGuide/using-eni.html

**Elastic IP addresses** [↗](#) [pp. 8 · 10](#)

What a fixed address means, and what it costs
docs.aws.amazon.com/AWSEC2/latest/UserGuide/elastic-ip-addresses-eip.html

**EC2 pricing** [↗](#) [p. 4](#)

Today's On-Demand · Reserved · Spot prices
aws.amazon.com/ec2/pricing

**Pricing Calculator** [↗](#) [p. 4](#)

Rerun this sheet's numbers for your own setup
calculator.aws

**Launching your first workload on Amazon EC2** [↗](#) [p. 1](#)

A Korean-language session that launches one instance from scratch. Official AWS Korea channel
www.youtube.com/watch?v=Mg6LLromiQM

**One to Many: Evolving VPC Design** [↗](#) [p. 11](#)

What breaks first when one server becomes many
aws.amazon.com/blogs/architecture/one-to-many-evolving-vpc-design

**Security groups** [↗](#) [pp. 9 · 10 · 13](#)

An allow-only firewall. Denying is the network ACL's job — coming next
docs.aws.amazon.com/AWSEC2/latest/UserGuide/ec2-security-groups.html

**EC2 Auto Scaling** [↗](#) [p. 4](#)

From one server to many, following demand — coming next
docs.aws.amazon.com/autoscaling/ec2/userguide/what-is-amazon-ec2-auto-scaling.html